

CSC-355 Homework #7

unification

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This homework is an opportunity for you to practice the steps in the PT algorithm.

Problem 1: PT Algorithm Terms

Give examples of the following things (make sure your examples are unique from the ones given in class):

- (a) a type expression
- (b) a type assignment
- (c) a type assertion that does not fail
- (d) a type assertion that does fail
- (e) an environment

Problem 2: PT Algorithm

Show the steps of the PT algorithm and what the algorithm returns (or how it fails to return) for the following inputs:

- (a) $\text{PT}(\lambda x.yx, \emptyset)$
- (b) $\text{PT}(\text{let } x = \lambda y.y \text{ in } x, \emptyset)$